Lecture 8 — Positive Definite Matrices

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Monday, October 5, 2026

After this lecture you should be able to

- State the definition of a positive definite matrix and read it as a statement about the quadratic form $\boldsymbol{x}^\mathsf{T}\boldsymbol{M}\boldsymbol{x}$.
- Recognise the bowl, the trough, and the saddle as the graphs of definite, semidefinite, and indefinite quadratic forms.
- Explain why positive definiteness is the multivariable second derivative test, and why we may restrict attention to symmetric matrices.
- Test a symmetric matrix for positive definiteness using its pivots or its eigenvalues.

Where we are

We now know how to take a symmetric matrix apart: real eigenvalues, orthogonal eigenvectors, $\boldsymbol{M} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\mathsf{T}$. Symmetry was the structural hypothesis. Today we add the one extra hypothesis that turns a symmetric matrix into something that behaves like a *positive number* — and that single extra property is what will later make energy positive, equilibrium a minimum, and the matrix $\boldsymbol{A}^\mathsf{T}\boldsymbol{C}\boldsymbol{A}$ invertible.

1 The definition

Definition 1.1 (Definiteness)**.** A square matrix $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ is **positive definite** (PD) if for all non-zero vectors $\boldsymbol{x} \in \mathbb{R}^n$,

$$\boldsymbol{x}^\mathsf{T}\boldsymbol{M}\boldsymbol{x} > 0.$$

$\boldsymbol{M}$ is **positive semi-definite** (PSD) if the inequality is not strict: for all non-zero $\boldsymbol{x} \in \mathbb{R}^n$ we have $\boldsymbol{x}^\mathsf{T}\boldsymbol{M}\boldsymbol{x} \geq 0$. If the inequality is flipped, $\boldsymbol{M}$ is said to be **negative definite**. If the inequality does not always hold in either direction, $\boldsymbol{M}$ is said to be **indefinite**.

Notice what kind of object $\boldsymbol{x}^\mathsf{T}\boldsymbol{M}\boldsymbol{x}$ is: $\boldsymbol{x}^\mathsf{T}$ is $1 \times n$, $\boldsymbol{M}$ is $n \times n$, $\boldsymbol{x}$ is $n \times 1$, so the product is 1×1 — a single number. The definition is a statement about the *sign* of that number, for every input at once.

Key idea

Positive definiteness is the generalisation of *positive numbers* to matrices. For $n = 1$ the matrix is a scalar m and $x\,m\,x = mx^2 > 0$ for all $x \neq 0$ says exactly $m > 0$. Everything we know about positive numbers — they have square roots, they are invertible, they make mx^2 a bowl — has a matrix analogue, and that analogue is this definition.

Definiteness is not a property of the entries

The definition quantifies over *all* non-zero $\boldsymbol{x}$. Testing a few convenient vectors, or glancing at the signs of the entries, proves nothing. The matrix $\boldsymbol{K}$ we have been using since Lecture 2 is full of -1's and is positive definite; we will check that at the end of the hour.

2 The geometrical picture: matrices as quadratic polynomials

A geometrical picture of PD matrices can be had by thinking of matrices as representing quadratic polynomials in n variables:

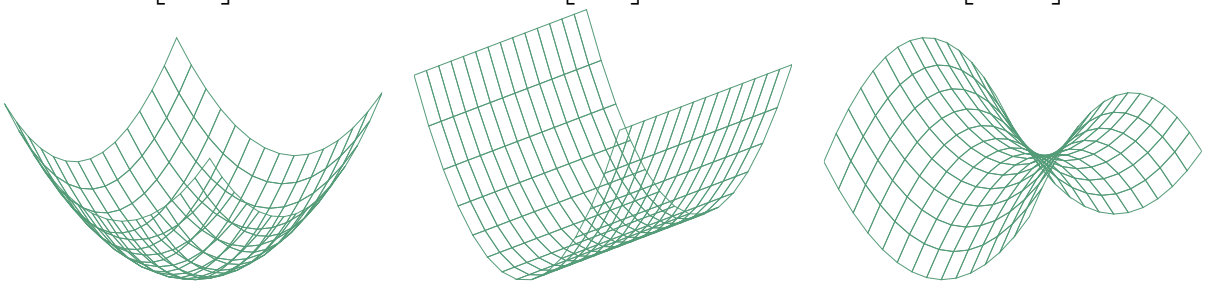
$$P_{\boldsymbol{M}}(x_1, x_2, \ldots, x_n) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}^{\mathsf{T}} \boldsymbol{M} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Every term of $P_{\boldsymbol{M}}$ is quadratic — entry m_{ij} contributes $m_{ij}x_ix_j$ — so $P_{\boldsymbol{M}}(\mathbf{0}) = 0$ always, and the whole question is what the graph does around the origin. That graph has different qualities depending on whether $\boldsymbol{M}$ is PD.

Visualisations for 2×2 matrices

Take the three simplest diagonal examples.

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$



$x^2 + y^2$ (definite) x^2 (semidefinite) $x^2 - y^2$ (indefinite)

A bowl, a trough, and a saddle. The trough is the interesting one: along the y-axis the polynomial x^2 never grows, so $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{M}\boldsymbol{x} = 0$ for a whole line of non-zero vectors. That is exactly the gap between > 0 and ≥ 0 in the definition.

Collecting the cases, the polynomial $P_{\boldsymbol{M}}(\boldsymbol{x})$ has

$P_{\boldsymbol{M}}$ **has**	**if** $\boldsymbol{M}$ **is**
a unique minimum	strictly positive definite
a subspace of minima	strictly positive semidefinite (but not PD)
a saddle point	indefinite
a subspace of maxima	strictly negative semidefinite
a unique maximum	strictly negative definite

Key idea

Positive definite = the quadratic form is a bowl with its single lowest point at the origin. Whenever we later find a physical energy that can be written as $\frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{M}\boldsymbol{x}$, positive definiteness of $\boldsymbol{M}$ is the statement that the system has one stable equilibrium and that any displacement costs energy.

3 This is the second derivative test

Recall, for a single variable function $f(x)$, the second derivative test for whether a critical point $f'(x_0) = 0$ is a minimum:

$$f''(x_0) > 0 \quad \text{is a sufficient condition for } x_0 \text{ to be a local minimum.}$$

We want the same statement in n variables. Both derivatives have to be generalised.

The generalisation of the first derivative is the **gradient**

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix},$$

a vector. The generalisation of the second derivative is the **Hessian**

$$\boldsymbol{H}(\boldsymbol{x}) = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_1 \partial x_n} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \cdots & & \vdots \\ \vdots & & & \\ \dfrac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & & \dfrac{\partial^2 f}{\partial x_n^2} \end{bmatrix},$$

a matrix — one second derivative for every pair of directions.

Theorem 3.1 (Second derivative test in n variables)**.** *If $\nabla f(\boldsymbol{x}_0) = \boldsymbol{0}$, then $\boldsymbol{H}(\boldsymbol{x}_0)$ being positive definite is a sufficient condition for $\boldsymbol{x}_0$ to be a local minimum of f.*

Compare this with the table above and the statement is not mysterious: near a critical point f looks like a constant plus the quadratic form of its Hessian, and we have just catalogued what those quadratic forms look like. A PD Hessian means a bowl; an indefinite Hessian means a saddle, which is neither a maximum nor a minimum.

Why this lecture is the hinge of the course

Property ② below — all eigenvalues positive — is stated in an engineering setting where $\boldsymbol{M}$ represents a physical system of some kind. That is where we are heading. In Lectures 12–13 every equilibrium problem in this course gets written as $\boldsymbol{A}^\mathsf{T}\boldsymbol{C}\boldsymbol{A}\,\boldsymbol{u} = \boldsymbol{f}$: $\boldsymbol{A}$ measures displacements, $\boldsymbol{C}$ holds material constants, $\boldsymbol{A}^\mathsf{T}$ balances forces. The reason that framework works at all is that $\boldsymbol{A}^\mathsf{T}\boldsymbol{C}\boldsymbol{A}$ comes out positive definite whenever $\boldsymbol{C}$ is positive and $\boldsymbol{A}$ has independent columns — so the energy $\frac{1}{2}\boldsymbol{u}^\mathsf{T}\boldsymbol{A}^\mathsf{T}\boldsymbol{C}\boldsymbol{A}\boldsymbol{u}$ is positive, the equilibrium is a genuine minimum, and the matrix is invertible. Today is where that guarantee is purchased.

4 Why we always assume symmetry

In general, when we speak about PD matrices, we restrict our attention to *symmetric* matrices. Simply because symmetric matrices have a lot of nice properties, and because we can always understand the definiteness properties of a non-symmetric matrix through its symmetric component.

Suppose $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is not symmetric. Then for any $\boldsymbol{x} \in \mathbb{R}^n$,

$$\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} = (\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x})^\mathsf{T} = \boldsymbol{x}^\mathsf{T}\boldsymbol{A}^\mathsf{T}\boldsymbol{x},$$

where the first step is legal because $\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}$ is just a number, and transposing a number changes nothing. Averaging the two expressions,

$$\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} = \tfrac{1}{2}(\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{x}^\mathsf{T}\boldsymbol{A}^\mathsf{T}\boldsymbol{x}) = \boldsymbol{x}^\mathsf{T} \underbrace{\frac{\boldsymbol{A}+\boldsymbol{A}^\mathsf{T}}{2}}_{\text{the symmetric component of } \boldsymbol{A}} \boldsymbol{x}.$$

Key idea

The quadratic form cannot see the antisymmetric part of a matrix at all. Definiteness of any $\boldsymbol{A}$ is the definiteness of $\frac{1}{2}(\boldsymbol{A}+\boldsymbol{A}^\mathsf{T})$, which is symmetric. So assuming symmetry costs us nothing, and buys us real eigenvalues and orthogonal eigenvectors from last lecture.

5 Two equivalent tests

The definition is useful theoretically, but sometimes clunky when we need to check that any particular matrix is PD — we cannot try every $\boldsymbol{x}$. Here are two necessary and sufficient (read: *equivalent*) properties of symmetric PD matrices.

Theorem 5.1 (Tests for positive definiteness)**.** *Let $\boldsymbol{M}$ be symmetric. Then*

① *$\boldsymbol{M}$ has all positive Gaussian elimination pivots:*

$$\boldsymbol{M} = \boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^\mathsf{T} \text{ is PD} \iff \boldsymbol{D} \text{ has all positive diagonal entries.}$$

② *$\boldsymbol{M}$ has all positive eigenvalues. (Remember that since $\boldsymbol{M}$ is symmetric, all eigenvalues are already real, so "positive" is a meaningful demand.)*

Both tests are cheap to state and very different in character.

- ① is the quickest check for a given random symmetric $\boldsymbol{M}$: run elimination, which we already know how to do from Lecture 3, and watch the pivots. Nothing else is required.
- ② is very useful and important, especially in an engineering setting where $\boldsymbol{M}$ represents a physical system of some kind. Eigenvalues are frequencies, decay rates, stiffnesses; their sign is a physical statement, not a bookkeeping one.

There are other equivalent characterisations of symmetric PD matrices, but they are more niche and not relevant for this class.

Reading the eigenvalue test off the definition

The two tests are not accidents. Writing $\boldsymbol{M} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\mathsf{T}$ from last lecture and setting $\boldsymbol{y} = \boldsymbol{Q}^\mathsf{T}\boldsymbol{x}$ turns the quadratic form into $\boldsymbol{x}^\mathsf{T}\boldsymbol{M}\boldsymbol{x} = \boldsymbol{y}^\mathsf{T}\boldsymbol{\Lambda}\boldsymbol{y} = \sum_i \lambda_i y_i^2$, a sum of squares with the eigenvalues as weights. And $\boldsymbol{M} = \boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^\mathsf{T}$ with $\boldsymbol{z} = \boldsymbol{L}^\mathsf{T}\boldsymbol{x}$ gives $\sum_i d_i z_i^2$ in the same way. Both tests say the same thing: the form is a weighted sum of squares, and it is positive exactly when the weights

are.

6 Check: our own matrices

Exercise 6.1. The fixed–fixed and fixed–free negative second derivative matrices are PD. Verify it:

$$\boldsymbol{K} = \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{bmatrix}, \qquad \boldsymbol{T} = \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 1 \end{bmatrix}.$$

Use test ①: eliminate and look at the pivots. Every one of them should come out positive.

That $\boldsymbol{K}$ and $\boldsymbol{T}$ pass the test is the reason the fixed–fixed and fixed–free rods of Lecture 2 had unique equilibrium positions. It is also worth noticing which matrix is missing from that list: the free–free matrix $\boldsymbol{B}$ of Lecture 2 is singular, and the constant vector that made it singular gives $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{x} = 0$ — so $\boldsymbol{B}$ is positive *semi*-definite but not positive definite. The trough in the picture above and the floating free–free rod are the same phenomenon.

Where this is going

Next time we leave linear algebra for an hour and derive the 1D heat equation from conservation of energy and Fourier's law of cooling. The second derivative $\partial^2 u/\partial x^2$ that shows up there is the continuous object behind $-\boldsymbol{K}$, and its two boundary conditions are the fixed and free ends we just tested. The positivity established today is what makes heat in that rod decay towards equilibrium rather than grow.

Reading

Strang, *Computational Science and Engineering*, § 1.6 (positive definite matrices); § 1.3 for the $\boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^{\mathsf{T}}$ pivots used in test ①.